
Feedback or Annealing?

A Controlled Study of Adaptive State Aggregation

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Abstract

Adaptive state aggregation replaces full-state Bellman updates with cheaper updates on groups of states with similar estimated values. A feedback rule anneals the group width, narrowing it as the Bellman residual shrinks, but any gain may come from the schedule rather than the residual signal. I separate these effects on a 9,261-state inventory control problem. On this instance, the feedback widths follow a roughly geometric path. I use the seed-0 feedback run to match an open-loop geometric schedule, then compare the two methods under three update mixes, 20 paired seeds, and five compute budgets. Feedback shows no consistent benefit. Among budget comparisons with unadjusted 95% bootstrap intervals that exclude zero, its largest advantage is 0.2% of the comparison error and its largest disadvantage is 5.7%. After 10,000 iterations, every interval contains zero and lies within the preplanned ± 0.02 negligible range. By contrast, at (1, 20), annealing reduces error by 22.7% after 10^5 backups but increases it by 10.4% after 3×10^6 . Full-state value iteration meets its 10^{-10} stopping tolerance in 4.20M backups; every 10,000-iteration aggregation run uses at least 9.2M. Evaluations should therefore compare feedback with fixed-width and matched open-loop schedules, measure policy quality, and use several compute budgets.

1 Introduction

State aggregation reduces the cost of dynamic programming by solving a smaller approximation to a Markov decision process (MDP) [1–3]. In Chen et al. [4], states with similar current values share a group; full-state updates alternate with cheaper updates that sample one state per group. A fixed group width ε trades cost against accuracy: wide groups are cheaper, while narrow groups reduce approximation error.

The Bellman residual measures the change from another full update. It suggests using wide groups when estimates are changing rapidly and narrower groups as they stabilize. I call this gradual narrowing annealing. However, comparison with a fixed width cannot distinguish the value of the residual signal from the value of annealing.

I compare feedback with a geometric schedule matched to the seed-0 feedback run. Once matched, the schedule is fixed and never reads the residual. At each update mix, it uses the same starting width, minimum width, and number of cycles to that minimum. The observed paths motivate the geometric comparison, while Proposition 1 states the contraction result that applies within consecutive full-state sweeps. Across five budgets, feedback adds no consistent benefit. Annealing has a larger effect, and its direction changes with the compute budget.

34 2 Related work

35 Classical aggregation assigns one value to each state group; its errors depend on within-group value
 36 variation [1, 3, 6]. Adaptive methods group by residual size [2], adjust partial membership [7], divide
 37 regions using value, policy, influence, or variance [8], or divide them as the value function changes
 38 [9]. I use Chen et al. [4], which groups states by current value and has asymptotic error bound
 39 $2\varepsilon/(1 - \gamma)$.

40 Residuals give error bounds [10], have been minimized directly [11], and guide state selection in
 41 prioritized sweeping [12, 13]. Because a small residual need not imply a good policy [14], I also
 42 measure policy loss. Here the residual range changes only group width, not membership rules, state
 43 priorities, or the fitting objective; I do not propose a new representation or convergence theorem.

44 3 Method

45 For a finite discounted cost MDP, the Bellman update T maps the current value function V to

$$(TV)(s) = \min_a \left[c(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right]. \quad (1)$$

46 The solver groups values $V(s)$ into nonempty intervals of width ε . A full-state iteration evaluates
 47 (1) for every state. A grouped iteration samples one state from each group, evaluates one Bellman
 48 update for that state, and uses the result to update the shared group value with step size $\alpha_t = 1/\sqrt{t}$,
 49 where t counts grouped updates. A schedule (ℓ_g, ℓ_a) repeats ℓ_g full-state iterations followed by ℓ_a
 50 grouped iterations. The groups are rebuilt before each grouped phase. Their values are copied back
 51 to individual states before the next full-state phase.

52 After each full-state sweep, I record the residual range $r_t = \max_s (TV_t - V_t)(s) - \min_s (TV_t - V_t)(s)$.
 53 The tested feedback rule is

$$\varepsilon_t^{\text{RESIDUAL}} = \max\{\varepsilon_{\min}, cr_t\}, \quad c = 0.084, \quad \varepsilon_{\min} = 0.05. \quad (2)$$

54 The choice $c = 0.084$ does not use an outcome from a stochastic grouped update. At the preplanned
 55 $(2, 5)$ mix, the first grouped phase follows two common full-state sweeps. They give $r = 5.93429$
 56 deterministically, so $0.084r = 0.49848 \approx 0.5$, matching the coarsest fixed baseline. The initial width
 57 is therefore known before the first grouped update. It changes with ℓ_g , giving $\varepsilon_0^{\text{RESIDUAL}} = 0.400$,
 58 0.499 , and 0.538 at $(5, 2)$, $(2, 5)$, and $(1, 20)$. The main control never reads r_t :

$$\varepsilon_j^{\text{GEOMETRIC}} = \max \left\{ \varepsilon_{\min}, \varepsilon_0^{\text{RESIDUAL}} \left(\frac{\varepsilon_{\min}}{\varepsilon_0^{\text{RESIDUAL}}} \right)^{j/(C-1)} \right\}, \quad (3)$$

59 over grouped phases j , with $(\varepsilon_0^{\text{RESIDUAL}}, C)$ set per mix to $(0.400, 7)$, $(0.499, 13)$, and $(0.538, 14)$: the
 60 starting width, the minimum, and the cycle at which the seed-0 RESIDUAL run reaches the minimum.
 61 Once these values are set, GEOMETRIC follows the same schedule for all 20 seeds and never reads
 62 the residual. In contrast, RESIDUAL uses the residual to choose each width. A method with fixed
 63 $\varepsilon = 0.05$ tests whether annealing is useful at all. The repository in Appendix F provides additional
 64 controls and complete intervals.

65 **Study design.** Before running the experiment, I specified the $(2, 5)$ mix, $c = 0.084$, the 20 ms
 66 primary comparison, the ± 0.02 equivalence range, and two geometric controls starting at 0.4985.
 67 Their durations were $C = 14$ for the main control and $C = 1,429$ for a slow control. I call these
 68 choices preplanned. I later added the $(5, 2)$ and $(1, 20)$ mixes, the fixed-backup budget ladder, and
 69 the matched geometric control used in Tables 1 and 2. For this matched control, I take C at each mix
 70 from the seed-0 RESIDUAL run. At $(2, 5)$ it differs from the main preplanned control by one cycle.
 71 Section 5 shows what happens when the preplanned parameters are copied to the added mixes.

72 **Why use a geometric comparison?** The observed feedback paths are close to geometric. A
 73 standard result explains the decay within each uninterrupted block of full-state sweeps, but does not
 74 determine the rate across the intervening grouped phases.

Proposition 1 (consecutive full-sweep contraction, 5, §6.6). *If two full-state sweeps are consecutive, their residual ranges satisfy $r_{t+1} \leq \gamma r_t$.*

This contraction result is standard [5, §6.6]; I do not claim it as new. Appendix A includes the proof to make the constants explicit. Grouped phases can increase the residual, so the proposition gives no contraction bound across a complete alternating cycle and does not predict C . In this experiment, geometric fits to the feedback widths have per-cycle factors 0.698, 0.829, and 0.829 at the three mixes. This observation motivates the open-loop control; the exact pilot duration C is measured rather than derived from contraction.

Inventory MDP, protocol, and metrics. A state is the inventory vector $q \in \{-10, \dots, 10\}^3$, giving $|\mathcal{S}| = 21^3 = 9,261$. Five choices of quote aggressiveness trade fill probability against spread, and the one-period cost combines spread revenue with correlated inventory risk (Appendix C). I use $\gamma = 0.95$ and rescale costs so $\|V^*\|_\infty = 100$. This is a small market-making-style control model, not a calibrated market simulator. Each method runs for 10,000 iterations at each mix. The MDP instance is fixed, so seeds 0–19 vary only which state is sampled within each group. The same seeds are used for all methods. The main accuracy measure is $\text{err}_\infty = \|V - V^*\|_\infty$, the largest absolute error across states. I also evaluate the policy chosen from V using $\|V^{\pi_V} - V^*\|_\infty$. Plots against computation use counted Bellman backups. Each full-state sweep counts $|\mathcal{S}|$ backups, and each grouped sweep t counts K_t , the number of occupied groups. This count favors aggregation because it omits the residual scan and the $|\mathcal{S}|$ state operations for each partition rebuild and group-to-state expansion. Paired differences are summarized by their median and a 95% percentile bootstrap interval based on 10,000 resamples. The intervals for the added budget ladder are not adjusted for multiple comparisons.

4 Results

Once the residual and geometric rules reach $\varepsilon_{\min}$ their error curves are visually indistinguishable at every mix (Figure 1, Appendix B). Table 1 shows the small numerical differences. At all three mixes, the 95% interval for residual minus matched geometric both contains zero and stays within ± 0.02 . The median differences are 1.7×10^{-7} , 5.1×10^{-7} , and -0.00201 . Thus, when the comparison has the same starting width, minimum width, and time to that minimum, using the residual does not improve the result after 10,000 iterations.

Value error and policy quality need not agree. At (2, 5), the residual and matched geometric methods have nearly identical value error. Their paired policy-loss difference is 1.2×10^{-6} , with a 95% CI of $[-0.000197, 0.00251]$, so the interval contains zero. At (1, 20), the residual method is worse than fixed-0.05 on both measures: its value error is 0.01250 higher (95% CI $[0.00745, 0.01561]$), and its policy loss is 0.03529 higher ($[0.03350, 0.03748]$). Feedback does not help on either measure here. Within each mix, the median computation at the end differs by at most 0.13%. For the residual method, rebuilding and expanding groups adds state operations equal to 39–96% of its counted-backup total. Full-state VI reaches its 10^{-10} stopping tolerance in 454 sweeps, or 4.20M counted backups. Each 10,000-iteration aggregation run uses 9.2–67.7M counted backups. Aggregation is therefore not competitive on this example; the experiment studies one approximation method rather than proposing a better solver.

Separating annealing from feedback. Results after 10,000 iterations hide annealing because each scheduled method has used $\varepsilon_{\min}$ for most of the run; at (5, 2), residual and fixed end only 3.8×10^{-8} apart. Table 2 instead compares four earlier budgets while widths still change. It reports relative differences because a ± 0.02 threshold is uninformative when errors are near 10.

Table 1: Results after 10,000 iterations over 20 paired seeds. R/G/F denote residual, matched geometric, and fixed $\varepsilon = 0.05$; the policy-loss and backup columns use the same order. Backups are reported in millions. The interval summarizes within-seed differences.

Mix	$\text{err}_\infty(\text{R})$	$\text{err}_\infty(\text{G})$	$\text{err}_\infty(\text{F})$	R–G (95% CI)	Policy loss R/G/F	Backups (M) R/G/F
(5, 2)	0.05001	0.05001	0.05001	$1.7 \times 10^{-7} [-2.8 \times 10^{-7}, 9.0 \times 10^{-7}]$	0.00210/0.00210/0.00210	67.685/67.684/67.686
(2, 5)	0.13882	0.13882	0.13792	$5.1 \times 10^{-7} [-9.6 \times 10^{-5}, 2.0 \times 10^{-5}]$	0.01518/0.01280/0.01280	30.214/30.206/30.195
(1, 20)	0.20343	0.20115	0.19093	$-0.00201 [-0.00320, 0.00541]$	0.05178/0.05178/0.01649	9.185/9.186/9.196

Table 2: Median paired value-error difference at fixed budgets, as a percentage of the comparison error; negative favors the first method. GEOMETRIC—FIXED isolates annealing and RESIDUAL—GEOMETRIC the added effect of feedback. * marks a nominal 95% bootstrap interval excluding zero; intervals are not multiplicity-adjusted. Details are in Appendix D.

Mix	Contrast	10^5	3×10^5	10^6	3×10^6	final
(5, 2)	GEOMETRIC—FIXED	−0.0	−5.5*	+0.1*	−8.3*	−0.0
(5, 2)	RESIDUAL—GEOMETRIC	+0.1	+0.0	−0.2*	+0.0	+0.0
(2, 5)	GEOMETRIC—FIXED	−2.2*	−8.2*	+6.0*	−6.5*	+0.6*
(2, 5)	RESIDUAL—GEOMETRIC	+0.0	+1.6*	−0.0	+0.0	+0.0
(1, 20)	GEOMETRIC—FIXED	−22.7*	−14.5*	−8.7*	+10.4*	+5.4*
(1, 20)	RESIDUAL—GEOMETRIC	+3.6*	+5.7*	+3.9*	−0.4	−1.0

Looking only at unadjusted intervals that exclude zero, feedback’s largest advantage is 0.2% of the comparison error and its largest disadvantage is 5.7%. Annealing changes error by about 23%: at (1, 20) matched geometric is 22.7% better than a fixed width after 10^5 backups but 10.4% worse after 3×10^6 . The schedule changes both the result and its direction, while feedback adds no consistent benefit. Large differences occur during rapid error reduction; exact percentages depend on which completed update the budget reaches.

The timing-based primary measure was inconclusive. At the preplanned 20 ms endpoint, the (2, 5) comparison gave a residual-minus-preplanned-geometric median of -0.0417 with 95% CI $[-0.0459, 0.0011]$, which is inconclusive under the specified equivalence rule. Repeated runs selected different iterations because error was falling rapidly and machine load varied.

5 What the negative result shows

The schedules must be matched on the right scale. A preplanned comparison must use the unit that triggers width changes, here grouped phases rather than total iterations. Copied unchanged to (1, 20), the slow geometric schedule never reaches its minimum and ends ten times worse; correcting only its cycle count removes the difference (Appendix E).

A mismatched schedule creates an apparent effect. Fixed at (0.4985, 14) for every mix, the main preplanned geometric control differs from residual at (5, 2) by 0.00632, 95% CI $[0.006319, 0.006327]$, with all 20 seeds favoring geometric. The mismatch causes this precise difference: geometric starts 25% wider and takes twice as long to reach the minimum. Matching both features reduces the difference to 1.7×10^{-7} , with an interval containing zero. Low simulation variance can therefore make a mismatched comparison look like a feedback effect.

6 Limitations and conclusion

One deterministic instance, residual statistic, minimum width, and feedback constant cannot support claims about other MDPs. The backup count omits group maintenance. Feedback may still help when regional residuals differ, the best minimum width is unknown, or computation ends before refinement does.

In this experiment, residual feedback does not beat the matched geometric schedule in any consistent way. Annealing does matter, changing error by about 23% at one matched budget, but its effect changes sign. Attributing an improvement to feedback requires a preplanned schedule matched on its starting width, minimum width, and duration, along with several compute budgets.

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184 A Proof of Proposition 1

185 The following standard result says that the Bellman operator shrinks differences when they are
 186 measured by their range [5, §6.6]. The proof is included to make the constants in the bound explicit.

187 Fix V, U and a state s , and let a^* attain the minimum in $(TU)(s)$. Evaluating the bracket of (1) at a^*
 188 rather than at the minimizer for V gives

$$\begin{aligned} (TV)(s) &\leq c(s, a^*) + \gamma \sum_{s'} P(s' | s, a^*) V(s') = (TU)(s) + \gamma \sum_{s'} P(s' | s, a^*) (V - U)(s') \\ &\leq (TU)(s) + \gamma \max_{s'} (V - U)(s'), \end{aligned}$$

189 and the symmetric argument, taking a^* to attain the minimum in $(TV)(s)$, gives $(TV)(s) \geq$
 190 $(TU)(s) + \gamma \min_{s'} (V - U)(s')$. Both bounds hold at every s , so $(TV - TU)(s)$ lies in $[\gamma \min(V -$
 191 $U), \gamma \max(V - U)]$. Defining $\text{span}(x) = \max(x) - \min(x)$, it follows that

$$\text{span}(TV - TU) \leq \gamma \text{span}(V - U).$$

192 During consecutive full-state sweeps $V_{t+1} = TV_t$ and $V_{t+2} = TV_{t+1}$, so $r_{t+1} = \text{span}(TV_{t+1} -$
 193 $V_{t+1}) = \text{span}(TV_{t+1} - TV_t) \leq \gamma \text{span}(V_{t+1} - V_t) = \gamma \text{span}(TV_t - V_t) = \gamma r_t$.

194 The proposition does not cover a grouped phase. Such a phase copies updates from representative
 195 states to their groups instead of applying T to every state, so it can increase the residual range. No
 196 contraction factor can therefore be multiplied across complete alternating cycles without additional
 197 assumptions. On this inventory instance, the geometric reference curve holds empirically before the
 198 width reaches its floor for all 20 seeds at all three mixes.

199 B Error over computation

Rules reaching $\epsilon_{\min}$ have similar error floors — inventory $N = 3, Q = 10, \gamma = 0.95$

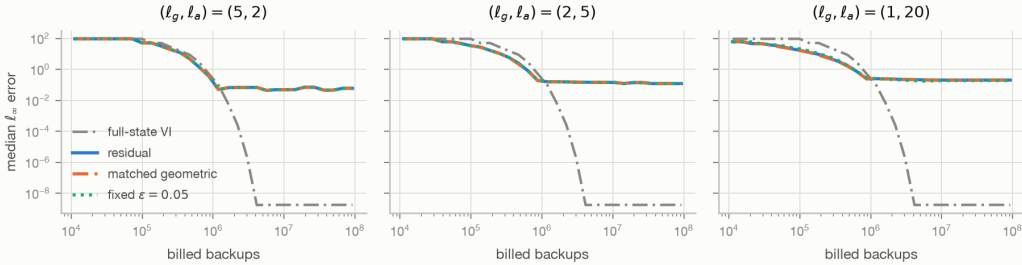


Figure 1: Median value error against counted Bellman backups. Final estimates are displayed beyond each method’s 10,000-iteration limit. Residual, matched geometric, and fixed-0.05 nearly coincide within each mix. The deterministic VI curve is repeated for reference; its convergence check stops after 454 sweeps (4.20M backups) though the plotted diagnostic continues to the common stopping point.

200 The flat right-hand portions repeat the final values; they do not represent continued computation. Each
 201 method stops after 67.7M, 30.2M, or 9.2M counted backups at (5, 2), (2, 5), or (1, 20), respectively.
 202 The plots carry the last value forward so that the three panels can share an axis. Budgets beyond a
 203 method’s stopping point therefore cannot be compared fairly, which is why the list in Appendix D
 204 stops at 3×10^6 .

205 C Inventory instance

206 The five joint quote-aggressiveness actions have fill probabilities (0.1, 0.2, 0.4, 0.6, 0.8) and spreads
 207 (1, 0.8, 0.5, 0.3, 0.15), so aggressive quotes fill often at a worse price. The one-period cost is

$$c(q, a) = 0.02 q^\top \Sigma q - \delta_a f_a L(q),$$

208 where δ_a and f_a are the spread and fill probability of action a , Σ has unit diagonal and correlation 0.5
209 so inventory risk is correlated across the three assets, and $L(q) = (6 - \#\{i : |q_i| = 10\})/6$ removes
210 fills that would leave the box $\{-10, \dots, 10\}^3$. Costs are rescaled after solving so that $\|V^*\|_\infty = 100$,
211 which puts ε , the equivalence margin, and every reported error on one interpretable scale.

212 **D How the budget comparisons are calculated**

213 Each cell of Table 2 is the median over 20 seeds of the within-seed difference divided by the
214 comparison method’s error for that seed at the same budget. Calculating the ratio before taking the
215 median preserves the pairing. Errors are read as a step function of counted backups: each budget
216 uses the most recent completed update. The list stops at 3×10^6 because the cheapest mix, $(1, 20)$,
217 performs only 9.2M backups in total. Later budgets would compare a repeated final value with
218 continued computation. The repository artifact reports absolute differences and 10,000-resample
219 percentile-bootstrap intervals for both the absolute and relative medians.

220 **E Adjusting the slow schedule across update mixes**

221 The slow preplanned geometric method used 1,429 grouped-update cycles, exactly the number
222 induced by 10,000 iterations at cycle length seven. Transplanted unchanged to $(1, 20)$, whose cycle
223 length is 21, it receives only 477 grouped phases and is still narrowing at the stopping point. It ends
224 at $\varepsilon = 0.232$ and error 1.59, an order of magnitude worse than every other method. Recalculating
225 only its cycle count to 477, changing nothing else, ends it at $\varepsilon = 0.05024$ and error 0.209, back in the
226 neighborhood of residual and matched geometric. The poor result came from an unfinished schedule,
227 not from the lack of feedback. For this reason, the added mixes use the schedule with the corrected
228 cycle count.

229 **F Code and result artifact**

230 The repository at <https://github.com/sunnysanitize/Residual-Epsilon-Aggregation>
231 provides code, configurations, complete intervals, and additional controls.